

Session 02 (AIML) -Assignment Question

- **Name of Student :** Shruti Dnyaneshwar Darwatkar
- **Domain Name:** AIML Development
- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.

PROGRAM:

```
#session 2
#Q1

# Ask the user to enter age
age = int(input("Enter your age: "))

# Check if age is 18 or more
if age >= 18:
    print("You are eligible to vote.")
```

OUTPUT:

```
Enter your age: 18
You are eligible to vote.
```

Q2.

PROGRAM:

```
#Q2

# Ask the user to enter a number
num = int(input("Enter a number: "))

# Check whether the number is even or odd
if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

OUTPUT:

```
Enter a number: 6
Even Number
```

```
Enter a number: 4
Even Number
```

Q3.

PROGRAM:

```
#Q3

marks = int(input("Enter marks: "))
if marks >= 90:
    print("Grade A - Excellent")
elif marks >= 75:
    print("Grade B - Good")
elif marks >= 60:
    print("Grade C - Average")
elif marks >= 40:
    print("Grade D - Pass")
else:
    print("Fail")
```

OUTPUT:

```
Enter marks: 77
Grade B - Good
```

```
Enter marks: 98
Grade A - Excellent
```

```
Enter marks: 65
Grade C - Average
```

```
Enter marks: 33
Fail
```

Q4.

PROGRAM:

Q4(a)

```
#Q4
#Q4(a). Print All Numbers from 1 to 30
## range(1, 31) generates numbers from 1 to 30

for i in range(1, 31):
    print(i)
|
```

OUTPUT:

```
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
```

Q4(b)

PROGRAM:

```
#Q4(b). Print All Odd Numbers from 1 to 50  
# range(1, 51, 2) generates odd numbers from 1 to 49  
  
for i in range(1, 51, 2):  
    print(i)
```



OUTPUT:

```
1  
3  
5  
7  
9  
11  
13  
15  
17  
19  
21  
23  
25  
27  
29  
31  
33  
35  
37  
39  
41  
43  
45  
47  
49
```

Q4(c)

PROGRAM:

```
#Q4(c). Print the Sum of Numbers from 1 to 15

# Calculate the sum of numbers from 1 to 15

total = 0

for i in range(1, 16):
    total += i

print("Sum =", total)
```



OUTPUT:

```
Sum = 120
```

Q5.

PROGRAM:

```
#Q5

i = 1
while i <= 20:
    print("Number =", i, "Cube =", i**3)
    i += 1
```

OUTPUT:

Number = 1 Cube = 1
Number = 2 Cube = 8
Number = 3 Cube = 27
Number = 4 Cube = 64
Number = 5 Cube = 125
Number = 6 Cube = 216
Number = 7 Cube = 343
Number = 8 Cube = 512
Number = 9 Cube = 729
Number = 10 Cube = 1000
Number = 11 Cube = 1331
Number = 12 Cube = 1728
Number = 13 Cube = 2197
Number = 14 Cube = 2744
Number = 15 Cube = 3375
Number = 16 Cube = 4096
Number = 17 Cube = 4913
Number = 18 Cube = 5832
Number = 19 Cube = 6859
Number = 20 Cube = 8000

Q6.

PROGRAM:

```
#Q6

total = 0

while True:
    num = int(input("Enter a positive number: "))

    if num <= 0:
        break

    total += num

print("Total Sum =", total)
```

OUTPUT:

```
Enter a positive number: 7
Enter a positive number: -4
Total Sum = 7
```


Q7.

PROGRAM:

```
#Q7 Weather advice using nested if

temperature = int(input("Enter temperature: "))
is_raining = input("Is it raining? (yes/no): ")

if temperature > 30:
    if is_raining == "yes":
        print("Hot and rainy, carry umbrella.")
    else:
        print("Hot day, wear light clothes.")

elif temperature < 15:
    if is_raining == "yes":
        print("Cold and rainy, wear jacket and take umbrella.")
    else:
        print("Cold day, wear warm clothes.")

else:
    print("Weather is pleasant.")
```

OUTPUT:

```
Enter temperature: 35
Is it raining? (yes/no): no
Hot day, wear light clothes.
```

Q8.

PROGRAM:

```
#Q8

for i in range(1, 41):

    if i % 3 == 0 and i % 5 == 0:
        print("FizzBuzz")

    elif i % 3 == 0:
        print("Fizz")

    elif i % 5 == 0:
        print("Buzz")

    else:
        print(i)
```

OUTPUT:

```
1
2
Fizz
4
Buzz
Fizz
7
8
Fizz
Buzz
11
Fizz
13
14
FizzBuzz
16
17
Fizz
19
Buzz
Fizz
22
23
Fizz
Buzz
26
Fizz
28
29
FizzBuzz
31
32
Fizz
34
Buzz
Fizz
37
38
Fizz
Buzz
```

Q9.

PROGRAM:

```
#Q9
while True:

    print("\n1. Add two numbers")
    print("2. Subtract two numbers")
    print("3. Multiply two numbers")
    print("4. Divide two numbers")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 5:
        print("Program Ended")
        break

    num1 = float(input("Enter first number: "))
    num2 = float(input("Enter second number: "))

    if choice == 1:
        print("Result =", num1 + num2)

    elif choice == 2:
        print("Result =", num1 - num2)

    elif choice == 3:
        print("Result =", num1 * num2)

    elif choice == 4:
        if num2 == 0:
            print("Division by zero is not allowed.")
        else:
            print("Result =", num1 / num2)

    else:
        print("Invalid Choice")
```

OUTPUT:

```
1. Add two numbers
2. Subtract two numbers
3. Multiply two numbers
4. Divide two numbers
5. Exit
Enter your choice: 7
Enter first number: 4
Enter second number: 2
Invalid Choice
```

```
1. Add two numbers
2. Subtract two numbers
3. Multiply two numbers
4. Divide two numbers
5. Exit
Enter your choice: 5
Program Ended
```

Q10.

PROGRAM:

```
#Q10

num = int(input("Enter a number: "))

if num > 100:
    print("Large number")

elif num > 50:
    print("Medium number")

else:
    print("Small number")

count = 1

while count <= 10:
    print(count)
    count += 1
```

OUTPUT:

```
Enter a number: 101
Large number
1
2
3
4
5
6
7
8
9
10
```